

Blood Bank Inventory & Emergency Donor Matcher

Domain: Healthcare & Telemedicine
Target Actors: Emergency Requester, Blood Bank Manager

1. Requirements Table

Functional Requirements

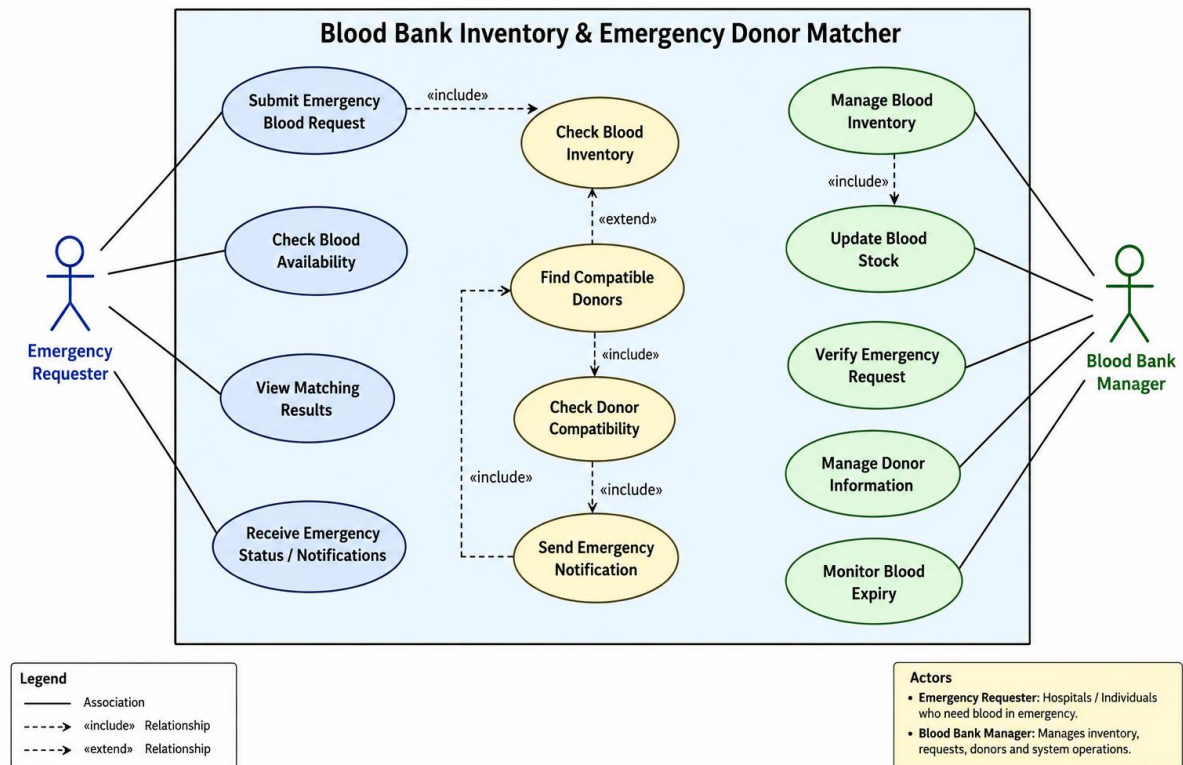
ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow an Emergency Requester to submit an emergency blood request by specifying blood group, blood component, required quantity, hospital/location, and urgency.	High	Pass: Request is successfully recorded with all mandatory details and an emergency status. Fail: Request is rejected if mandatory information is missing.	Enables hospitals or emergency requesters to quickly communicate urgent blood requirements.
FR-002	Functional	The system shall cross-reference emergency blood requests with real-time blood bank inventory and identify compatible available blood units.	High	Pass: System displays compatible blood units and their quantities. Fail: System reports that no compatible inventory is available.	Helps locate available blood immediately and reduces delays during emergencies.
FR-003	Functional	The system shall identify eligible blood donors within a configurable geographical radius based on blood-group compatibility and donor availability.	High	Pass: Eligible compatible donors within the specified radius are identified. Fail: No incompatible donor is selected.	Helps locate nearby donors when blood bank inventory is insufficient.
FR-004	Functional	The system shall send emergency notifications to eligible compatible donors when a critical blood shortage is detected.	High	Pass: Compatible donors receive an emergency notification containing the required blood group, location, and urgency. Fail: Incompatible or	Enables rapid donor response during critical shortages.

ID	Type	Description	Priority	Acceptance Criteria	Rationale
				unavailable donors are not notified.	
FR-005	Functional	The system shall allow the Blood Bank Manager to add, update, reserve, and remove blood units while maintaining their blood group, component type, quantity, expiry date, and availability status.	High	Pass: Inventory changes are reflected immediately and correctly. Fail: Invalid or conflicting updates are rejected.	Maintains accurate real-time blood stock information and prevents allocation errors.

Non-Functional Requirements

ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Performance & Security	The system shall maintain transactional consistency and prevent simultaneous allocation of the same blood unit while protecting sensitive donor and patient information through authenticated and authorized access.	High	Pass: Concurrent allocation tests never assign the same blood unit twice, and unauthorized users cannot access protected information. Fail: Duplicate allocation or unauthorized access occurs.	Prevents critical inventory errors and protects sensitive healthcare information.
NFR-002	Performance & Availability	The system shall process emergency matching and notification requests with low latency and remain available during high-demand emergency situations.	High	Pass: Emergency matching results are generated within the defined target response time under simulated peak load. Fail: Response time exceeds the target or the service becomes unavailable under expected load.	Emergency blood requests are time-critical and require a responsive and reliable system.

2. UML Use-Case Diagram



3. Use-Case Flow Specification

Use Case: Emergency Blood Request & Donor Matching

Use Case ID: UC-01

Use Case Name: Emergency Blood Request & Donor Matching

Primary Actor

Emergency Requester

Supporting Actor

Blood Bank Manager

Goal

To obtain compatible blood units quickly from available blood-bank inventory or, when inventory is insufficient, identify and notify eligible nearby donors.

Preconditions

1. The Emergency Requester has access to the system.

2. The requester has valid emergency blood requirement information.
 3. Blood inventory and donor information are available in the system.
 4. The system is operational and connected to the notification service.
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Postconditions

Success

1. The emergency request is recorded.
2. Compatible blood inventory is identified and reserved if available.
3. If inventory is insufficient, compatible nearby donors are identified.
4. Emergency notifications are sent to eligible donors.
5. The requester receives the matching status.

Failure

1. The emergency request remains recorded.
 2. The requester is informed if no compatible blood or eligible donors are available.
 3. No incompatible blood unit or donor is allocated.
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Main Success Scenario

1. **Emergency Requester** logs into the system.
 2. The requester selects **Submit Emergency Blood Request**.
 3. The requester enters the required blood group, blood component, quantity, hospital/location, and urgency.
 4. The system validates the submitted information.
 5. The system records the emergency request.
 6. The system checks the real-time blood-bank inventory.
 7. The system identifies compatible blood units.
 8. If sufficient compatible blood is available, the system reserves the required units.
 9. The system updates the inventory to reflect the reservation.
 10. The system displays the matching result to the Emergency Requester.
 11. The Blood Bank Manager can verify and process the emergency request.
 12. The use case ends successfully.
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Alternate Flow — Insufficient Blood Inventory

At Step 8:

1. The system determines that the available compatible blood inventory is insufficient.
2. The system activates the **Find Compatible Donors** use case.

3. The system checks donor blood-group compatibility and eligibility.
 4. The system filters donors based on the configured geographical radius.
 5. The system identifies eligible nearby donors.
 6. The system sends emergency notifications to the identified donors.
 7. The requester is informed that donor matching has been initiated.
 8. The system records the notification and matching status.
 9. The use case ends with the emergency request remaining active until sufficient blood is obtained.
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Exception Handling

If no compatible blood units and no eligible donors are found, the system shall:

1. Inform the Emergency Requester that no immediate match is available.
 2. Keep the emergency request active.
 3. Notify the Blood Bank Manager.
 4. Continue allowing the Blood Bank Manager to update inventory or donor information.
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